

$m(t)$ is sh

Question # 6 [CLO: 8]

[10]

Identify and mention the type of digital band pass modulation techniques shown in figure 4. Also write the general analytical expression for each waveform.

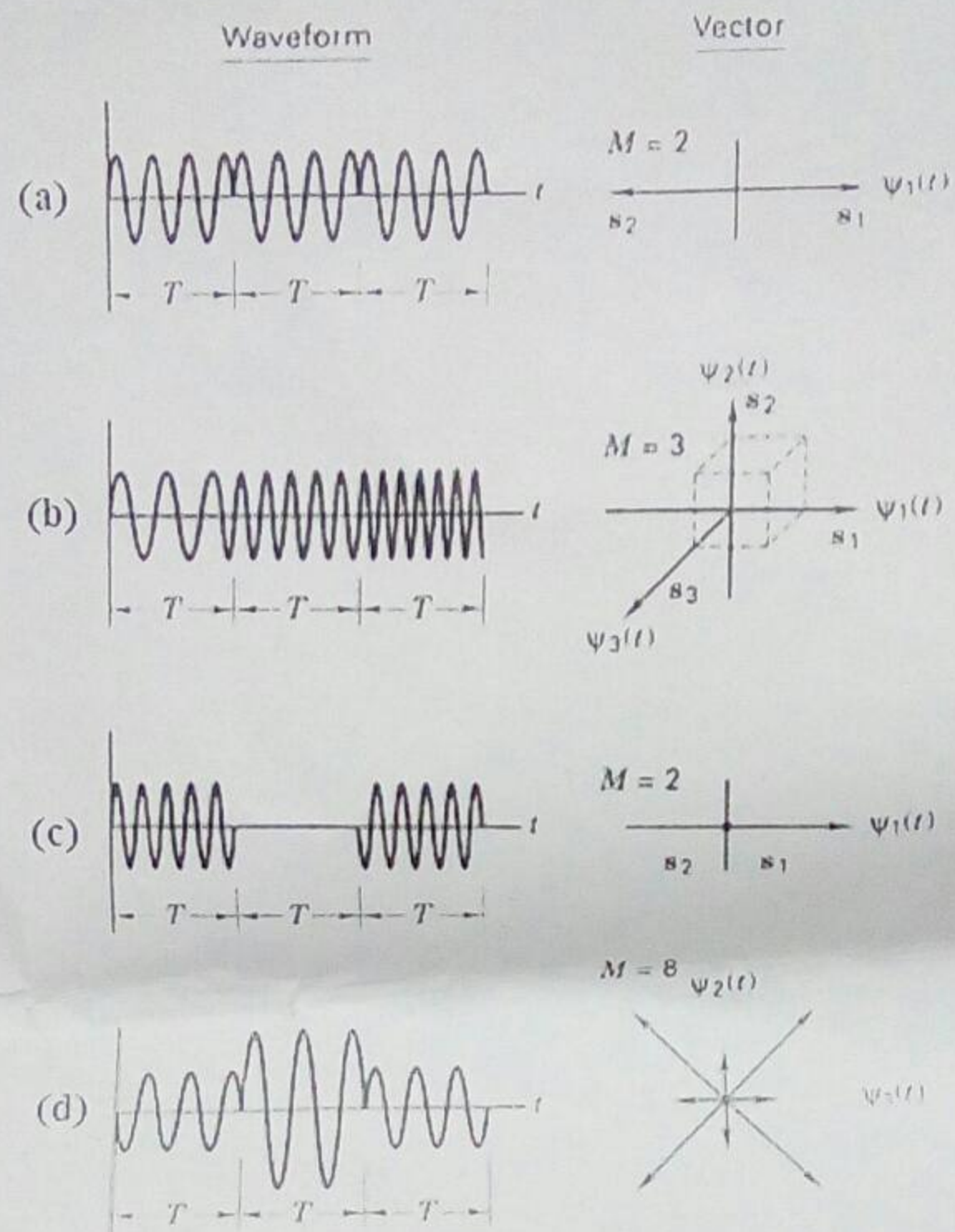


Figure 4

National University of Computer and Emerging Sciences, Lahore Campus



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Instruction/Notes: Attempt all parts of a question together.
Fourier Transforms Table is provided.
Attach the question paper with answer sheet.

Question # 1 [CLO: 2]

[20]

A random binary pulse train $g(t)$ is shown in figure 1. The pulse width is $T_b/4$, and one binary digit is transmitted every T_b seconds. A binary 1 is transmitted by the positive pulse, and a binary 0 is transmitted by the negative pulse. The two symbols are equally likely and occur randomly.

- Find the autocorrelation function $R_g(\tau)$ and plot it.
- Find the power spectral density $S_g(\omega)$ and plot it.
- Estimate the essential bandwidth of the signal.

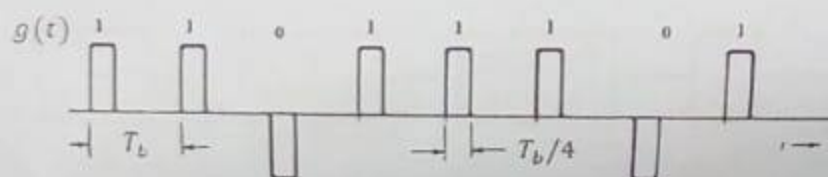


Figure 1

Question # 2 [CLO: 3 & 4]

[20]

A double sideband with carrier (DSB-WC/AM) signal is given by

$$\phi(t) = 5 \cos 1800\pi t + 20 \cos 2000\pi t + 5 \cos 2200\pi t$$

- Sketch the spectrum of $\phi(t)$ and determine the bandwidth.
- Determine the modulating signal $m(t)$ and the carrier signal $c(t)$. Write the equations.
- Determine the modulation index.
- Determine the power efficiency.
- Determine and sketch the $SSB_{LSB} - WC$ signal through filtering method. Clearly specify filter parameters.
- Can we demodulate the $SSB - WC$ signal found in part 'e' using coherent technique? To support your answer provide the block diagram along with mathematical and graphical proofs.

Question # 3 [CLO: 7]

[10]

Show that Delta modulation is a special case of Differential Pulse Code Modulation. Also show the practical implementation of delta modulator and demodulator. To support your answer provide the block diagram and mathematical proof wherever required.

Question # 4 [CLO: 5]

Consider the PM modulated signal $\varphi_{PM}(t) = 10 \cos(2\pi \cdot 10^5 t + k_p m(t))$ where $k_p = 2\pi$ and $m(t)$ is shown in figure 2. Assume $B = 1 \text{ kHz}$.

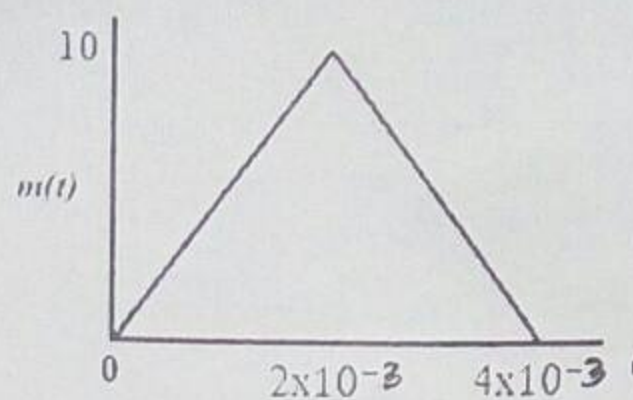


Figure 2

- Find the power of modulated signal.
- Find the phase deviation $\Delta\phi$.
- Find the frequency deviation Δf .
- Estimate the bandwidth of $\varphi_{PM}(t)$.
- Sketch the PM waveform
- Can we demodulate $\varphi_{PM}(t)$ using demodulator shown in figure 3? To support your answer provide proper reasoning and mathematical proof. Suggest modification for given demodulator if required?

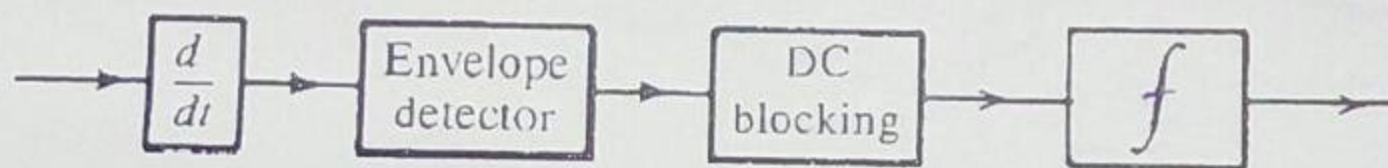


Figure 3

Question # 5 [CLO: 7]

[20= 5+15]

- Explain the problem of "Aliasing" with the help of a figure and discuss its solution.
- A television signal has a bandwidth of 4.5 MHz. This signal is sampled, quantized, and binary coded to obtain a PCM signal. Signal must be sampled at a rate 25% above the Nyquist rate and quantized into 4096 levels. The signal has average power of 0.1 Watt and peak voltage of 1 volt.
 - Determine the sampling rate.
 - Determine the minimum number of bits per second required to encode the signal.
 - Determine the minimum bandwidth required to transmit this signal.
 - Determine the SNR for PCM case.
 - If the television signal is transmitted using DM technique, what would be the bandwidth required to transmit this signal, when signal is sampled at rate of 40 MHz.